ShopInsight AI

E-Commerce Purchase Intelligence & Decision-Support System

CMPT 310 - Artificial Intelligence | Project Proposal

1. Group Information

Group	ShopInsight AI (Project Group 13)
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2. Project Idea

Problem & AI Task

Our project, **Online Shopping Purchase Prediction**, will develop an AI system that predicts whether an online shopping session will result in a purchase based on the user's activity during the session. This is a **supervised binary classification task**. The system will learn from previous shopping sessions with known outcomes and classify a new session as either **Purchase** or **No Purchase**.

System Input & Output

The **input** to the system will be information describing an online shopping session. This includes features such as the number of pages visited, time spent on different types of pages, visitor type, month, weekend status, and other session-related features provided in the dataset. The **output** will be a binary classification:

- **Purchase** - the session is predicted to result in a purchase.
- **No Purchase** - the session is predicted not to result in a purchase.

AI Techniques:

The main AI approach will be **supervised machine learning using classification algorithms**. We plan to begin with **Logistic Regression** as a baseline model and compare it with other classification approaches such as a **Decision Tree** and **K-Nearest Neighbours (KNN)**. We also plan to experiment with a small Neural Network after completing the baseline models. The models will be evaluated using metrics such as **accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC**. Comparing several models will help us determine which approach performs best for predicting whether an online shopping session will result in a purchase.

Proposed System Flow



Figure 1. Proposed AI pipeline.

3. Tools and Resources

Python will be used to develop the system, **scikit-learn** for preprocessing and classification models, **pandas** for data preparation, and **Matplotlib** for result visualization.

Dataset:

We plan to use the **Online Shoppers Purchasing Intention Dataset from the UCI Machine Learning Repository** [1]. It contains 12,330 online shopping sessions with features such as page activity, time spent on different page types, visitor type, month, and weekend status. The variable Revenue indicates whether a session resulted in a "purchase" and will be used as the classification label. The dataset will be divided into training and testing data, so that the models can be evaluated on unseen shopping sessions (the testing set).

Own dataset/simulator:

We do not plan to generate our own dataset or use a simulator

We will work with the existing real-world session data provided by UCI

4. Project Plan / Timeline

Stage / Target	AI & Technical Work	Working Deliverable	Success Check
Milestone 1 - November 1	Audit features and class balance; build train/validation/test workflow; encode categorical variables; scale where required; fit preprocessing only on training data; train logistic baseline plus at least one comparison model.	Reproducible end-to-end pipeline that accepts session features and returns purchase/no-purchase plus probability; initial model comparison and plots.	Initial accuracy, precision, recall, F1, confusion matrix and ROC-AUC reported; pipeline runs from raw data to prediction.
Milestone 2 - November 29	Train/tune neural network; study SGD/backpropagation, regularization and model complexity; compare models under the same protocol; tune hyperparameters/threshold on validation data; reserve test data for final evaluation.	Selected final model; explainability view; low/medium/high intent output; model-comparison visualizations; integrated demo; documented limitations, ethics and reproducibility.	Final held-out evaluation completed; minority-class performance examined; training-vs-validation behaviour checked; complete system can be demonstrated reliably.
Optional Stretch After core works	Explore unsupervised clustering for behavioural session groups and a what-if sensitivity interface that recalculates model predictions when inputs change.	Customer-segment exploration and interactive sensitivity view added only if MVS and Milestone 2 requirements are stable.	Stretch work remains clearly separated from causal claims: it explains model associations, not guaranteed business outcomes.

5. Minimal Viable System

Simplest Working Version

A Python program using **Logistic Regression** trained on the UCI dataset to predict **Purchase or No Purchase** for a shopping session, along with its purchase probability.

Core Functionality We Will Prioritize First

The core functionality will be a reliable pipeline for **data preprocessing, Logistic Regression training, purchase prediction, and evaluation** before adding more advanced models.

Usefulness in Basic Form

Even this basic version can identify sessions with higher predicted purchase intent and reveal browsing patterns associated with purchases. It will also provide an interpretable baseline for evaluating more advanced models.

References

[1] Sakar, C., & Kastro, Y. (2018). Online Shoppers Purchasing Intention Dataset. UCI Machine Learning Repository. DOI: 10.24432/C5F88Q. Official UCI dataset page